A Structured Research Journey: A Synthetic Example Thesis

A thesis submitted in fulfilment of the requirements for the degree of

Doctor of Philosophy

in

Your Discipline

by

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Your School

Your Faculty

Your University

Month 20XX

Certificate of Original Authorship

I, *Student Name*, *[Replace this example: insert the exact current Certificate of Original Authorship wording required by your university. Do not submit this placeholder.]*

[Replace this example: Include every declaration, disclosure and acknowledgement required for your degree, school and circumstances. Requirements may change over time.]

Signature: ______________________________

Date: ____________________ Place: ____________________

Generative AI disclosure example

Important for UTS graduate research students. At the time this example was checked, the UTS Graduate Research Candidature Management, Thesis Preparation and Submission Procedures (version 1.19) required every graduate research student to indicate how generative AI was used in the research or preparation of the thesis. The selected statement forms part of the Certificate of Original Authorship. Students reporting substantive use must also provide the required usage table.

Do not submit this community example as official wording. Download the current UTS Thesis Preparation and Submission Procedures, choose the statement that truthfully describes your use, and follow instructions from your supervisor, faculty and Graduate Research School. Requirements may change.

Choose the truthful usage category or categories

No substantive use: no deliberate AI use beyond features unavoidably built into ordinary software or search tools.

Assistive use: help with the researcher's own expression or workflow, such as organisation, grammar, brainstorming or LaTeX troubleshooting, without supplying original ideas, interpretations or data.

Generative use: production of text, images, code, summaries or other material that may enter the research output, even when the researcher later rewrites it.

Analytical use: interpretation, evaluation, classification, critique, pattern identification or suggested conclusions concerning data or research material.

Using Codex, Claude, ChatGPT or another system deliberately for editing, drafting, coding, summarising or analysis is not consistent with choosing "no substantive use". Students must classify what they actually did and use their university's current exact statement.

Synthetic example for a student who used AI

[Replace this example: Replace the following paragraph and every table row with a complete and truthful account. Delete the table only when the current official procedure permits it.]

Generative AI tools were used in an *[Replace this example: assistive, generative and/or analytical]* capacity. The candidate remains responsible for the intellectual contribution, originality, accuracy, research integrity and final wording. All outputs were reviewed and verified before incorporation. No confidential research data or identifiable participant information was provided to an unapproved service.

Chapter or location	**Tool, version and platform**	**Purpose and how outputs were incorporated**
Throughout thesis preparation	*[Replace this example: Provider and tool; model/version; VS Code, web, app or API; dates used]*	*[Replace this example: For example: LaTeX build diagnosis and file organisation. State what was accepted, rewritten, rejected and independently checked.]*
[Replace this example: Chapter or section]	*[Replace this example: Tool, model/version, platform and date]*	*[Replace this example: Exact task, AI-use category, incorporation and human verification.]*

Responsible-use and record-keeping checklist

Before signing the certificate, the student should be able to confirm that they have:

- recorded the tool/provider, model or version, platform, date, purpose, relevant prompts or queries, outputs, follow-up interactions and affected thesis locations;
- verified every factual statement, number, quotation and citation against an authoritative original source, and checked for hallucination, plagiarism and bias;
- retained human authorship and responsibility for research design, analysis, interpretation, conclusions and the submitted wording;
- complied with ethics approvals, consent, confidentiality, data classification, intellectual-property, licence, contract and cybersecurity obligations;

- avoided uploading participant data, confidential examiner material, restricted research, unpublished third-party work or university information to an unapproved tool; and

- followed current university, faculty, school, supervisor, funder and publisher requirements for permission, acknowledgement and disclosure.

UTS students must also consult the current Use of AI in Research Guidelines. The guidance emphasises documentation, human accountability, verification, privacy, confidentiality, appropriate tool selection, research integrity and disclosure.

Dedication

[Replace this example: Write a short dedication, or remove this page.]

Acknowledgements

[Replace this example: Thank supervisors, collaborators, participants, funders, family and community members as appropriate. Check whether funding bodies require exact wording.]

Example: I sincerely thank my supervisory panel for their guidance throughout this research. I am also grateful to the participants who generously contributed their time and experience. Any remaining errors are my own.

Abstract

[Replace this example: Replace this example with a self-contained summary of the problem, method, principal results, contribution and implications. Use verified numbers.]

Example structure: This thesis investigates a clearly defined research problem within a specified context. It uses a transparent research design to collect and analyse relevant evidence. The findings identify an important pattern and explain its significance for theory and practice. The thesis contributes a reproducible approach and identifies limitations that should guide future research.
Keywords: example keyword; research method; thesis template; replace these terms

Thesis Format

[Replace this example: State whether the thesis is a conventional monograph, thesis by publication, creative work with exegesis, or another approved format.]

Example: This thesis is presented as a conventional monograph. Chapters 1–3 establish the research problem and methodology, Chapters 4–5 report and discuss the findings, and Chapter 6 presents the conclusions and recommendations.

Publications and Presentations

[Replace this example: List only verified outputs and explain how each relates to the thesis.]

Example:

- Student, A. and Supervisor, B. (20XX). “Example publication title.” *Example Journal*. Status: published / accepted / under review.

Statement of Author Contributions

[Replace this example: For thesis-by-publication content, state each contributor's role and use the exact authorship requirements supplied by your school or university.]

Example: The student led the study design, data analysis and manuscript drafting. The supervisory panel contributed methodological guidance, critical revision and research oversight.

Contents

Certificate of Original Authorship i

Dedication v

Acknowledgements vi

Abstract vii

Thesis Format viii

Publications and Presentations ix

Statement of Author Contributions x

List of Figures xiii

List of Tables xiv

Abbreviations xv

1 Introduction 1

1.1 Background and context 1

1.1.1 Moving from a broad setting to a precise focus 1

1.2 Problem statement and research gap 2

1.3 Aim, objectives and research questions 2

1.4 Illustrative contributions 2

1.5 Thesis structure 3

2 Literature Review 4

2.1 Purpose and scope 4

2.2 Search and selection approach 4

2.3 Theme one: structured scholarly writing 4
2.4 Theme two: transparent revision 5
2.5 Synthesis and conceptual framework 5

3 Methodology **6**
3.1 Research design 6
3.1.1 Design rationale 6
3.2 Study workflow 6
3.3 Data collection or evidence generation 7
3.4 Analysis 7
3.5 Ethics, integrity and data management 7

4 Results **8**
4.1 Chapter overview 8
4.2 Example descriptive results 8
4.3 Example analytical result 9

5 Discussion **10**
5.1 Overview of principal findings 10
5.2 Interpretation in relation to prior work 10
5.3 Theoretical and practical implications 10
5.4 Strengths and limitations 11

6 Conclusion **12**
6.1 Research questions revisited 12
6.2 Contributions 12
6.3 Recommendations and future research 12
6.4 Closing statement 13

A Example Research Instrument **14**
A.1 Illustrative prompts 14

Bibliography **15**

List of Figures

3.1 Example research workflow drawn directly in LaTeX. 7

List of Tables

1.1 Example contribution map 3

2.1 Example literature synthesis table 5

4.1 Synthetic example results (replace completely) 8

Abbreviations

HDR Higher Degree by Research

AI Artificial Intelligence

UTS University of Technology Sydney

[Replace this example: Remove unused examples and add only abbreviations used in your thesis.]

CHAPTER 1

INTRODUCTION

[Replace this example: Rename this chapter and replace every example paragraph with your own work.]

1.1 Background and context

Start broadly enough for a reader outside your immediate specialisation. Define the real-world or scholarly setting, introduce the central concepts, and support factual claims with authoritative sources. For example, research projects often become harder to maintain when their writing, evidence, figures and revision history are scattered across unrelated files (Knuth, 1984).

1.1.1 Moving from a broad setting to a precise focus

A subsection develops one part of the chapter argument without creating a new chapter. This fully synthetic example demonstrates the standard hierarchy automatically: Chapter 1 contains Section 1.1, which contains Subsection 1.1.1. Students should use headings only when they help a reader follow the logic of the real thesis.

1.2 Problem statement and research gap

State what is known, what remains unresolved, and why that unresolved issue matters. Avoid presenting a broad topic as though it were automatically a research gap.

Although prior work explains how structured authoring tools support reproducible documents, there remains limited practical guidance connecting large-thesis file organisation, transparent examiner revisions and multiple submission outputs in one maintainable workflow. This thesis addresses that operational gap.

1.3 Aim, objectives and research questions

The aim should describe the overall intellectual destination. Objectives are the concrete steps used to reach it, while research questions define what the thesis will answer. A student could adapt the following pattern:

> **Aim:** To investigate how *[Replace this example: phenomenon]* affects *[Replace this example: population or system]* in *[Replace this example: context]*.

1. Review and synthesise the relevant literature.
2. Design and apply an appropriate research method.
3. Analyse the evidence and explain its implications.

RQ1 How does *[Replace this example: factor A]* influence *[Replace this example: outcome B]*?

RQ2 Under what conditions does that relationship change?

1.4 Illustrative contributions

Do not claim novelty merely because a technique is new to your project. Connect every claimed contribution to evidence developed in later chapters.

Table 1.1: Example contribution map

Contribution	**Claim to be demonstrated**	**Evidence location**
C1	A clearly defined conceptual or empirical contribution.	Chapters 2 and 4
C2	A methodological or practical contribution.	Chapters 3 and 5
C3	A set of implications grounded in the study evidence.	Chapters 5 and 6

1.5 Thesis structure

Chapter 2 synthesises prior research. Chapter 3 explains the research design, Chapter 4 reports the evidence, and Chapter 5 interprets it. Chapter 6 answers the research questions and identifies future work.

Chapter 2

Literature Review

2.1 Purpose and scope

Explain what the review covers, its boundaries and how it supports the research questions. A literature review should synthesise relationships and disagreements, not become a sequence of disconnected paper summaries.

2.2 Search and selection approach

Report databases, search concepts, date ranges, inclusion criteria and the final screening process at a level another researcher could understand. If the thesis is not a formal systematic review, describe the approach accurately rather than adopting a label that the method cannot support.

2.3 Theme one: structured scholarly writing

Structured source documents separate content from presentation and allow the same material to be transformed consistently (Lamport, 1994). The practical value for a large thesis is not simply attractive typesetting: stable labels, bibliographic automation and modular files reduce avoidable maintenance work.

2.4 Theme two: transparent revision

Revision evidence should connect a specific examiner comment to a specific substantive change. Colour can assist navigation, but identifiers such as `E1-01` preserve meaning when a document is printed in greyscale or read with assistive technology.

2.5 Synthesis and conceptual framework

Compare the themes, explain their relationship, and end with the precise proposition or gap taken into the methodology. Table 2.1 demonstrates a compact synthesis table.

Table 2.1: Example literature synthesis table

Theme	**What is established**	**What is contested or missing**	**Relevance to this thesis**
Structured writing	Modular sources improve document maintainability.	Practical workflows vary across disciplines.	Motivates the file architecture.
Revision evidence	Traceable responses improve review transparency.	Colour-only systems create accessibility risks.	Motivates IDs plus colour.

CHAPTER 3

METHODOLOGY

3.1 Research design

Name and justify the research design in relation to the research questions. Explain why it is a better fit than plausible alternatives, and distinguish the overall methodology from individual data-collection methods.

3.1.1 Design rationale

Use this level to explain a coherent component of the research design. A useful rationale connects the research question, the evidence required, the selected method and the limitations introduced by that choice. This text is synthetic guidance rather than a claim about a completed study.

3.2 Study workflow

As illustrated in Figure 3.1, the research questions determine the evidence required, the analysis converts that evidence into findings, and validation checks whether the resulting interpretation is sufficiently supported. The arrows represent an iterative process rather than an irreversible pipeline.

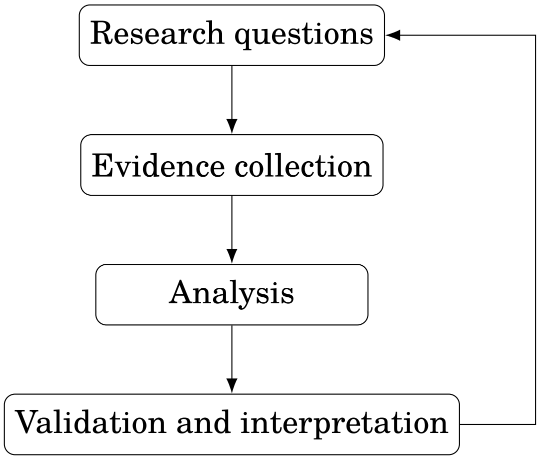

Figure 3.1: Example research workflow drawn directly in LaTeX.

3.3 Data collection or evidence generation

Describe who or what supplied the evidence, how items were selected, the procedure, instrument quality and any departures from the approved protocol. Replace this example with enough detail for a knowledgeable reader to assess validity and reproducibility.

3.4 Analysis

Define every measure and connect it to the question it answers. For example, an illustrative normalised change can be expressed as

$$\Delta_i = \frac{x_{i,\text{post}} - x_{i,\text{pre}}}{\max(x_{i,\text{pre}}, \epsilon)}, \tag{3.1}$$

where $x_{i,\text{pre}}$ and $x_{i,\text{post}}$ are the verified measurements for case i, and $\epsilon > 0$ prevents division by zero. Equation (3.1) is an example only; students must use an analysis justified for their own evidence.

3.5 Ethics, integrity and data management

State the relevant approval identifier, consent arrangements, privacy safeguards, storage and retention controls. Never copy an example approval number. Follow the current instructions from your institution, faculty or school and research ethics office.

CHAPTER 4

Results

4.1 Chapter overview

Report findings in the order needed to answer the research questions. Keep observation separate from interpretation when that distinction helps the reader; the discussion chapter can then explain meaning, implications and relationship to prior work.

4.2 Example descriptive results

The deliberately small dataset below shows how a student can combine a caption, stable label, numeric alignment and an explanatory note without converting the table to an image.

Table 4.1: Synthetic example results (replace completely)

Group	n	**Mean**	**SD**
Example A	24	4.20	0.61
Example B	26	4.68	0.57

Note. All values are invented solely to demonstrate table formatting.

The synthetic values in Table 4.1 demonstrate formatting only. They do not represent real participants, experiments, findings, people or institutions.

4.3 Example analytical result

A useful results paragraph identifies the comparison, reports the estimate and its uncertainty, and then states the factual result without exaggerating what the design can establish. Replace this paragraph with verified output from your own analysis.

These illustrative results cannot support scientific or institutional conclusions because the values are deliberately synthetic. In a real thesis, the limitations must address sampling, measurement, analytical assumptions, missing evidence and the boundaries of generalisation.

Chapter 5

Discussion

5.1 Overview of principal findings

Begin by answering the research questions in plain academic language. Do not repeat the entire results chapter; identify the most consequential findings and their relationship to the thesis aim.

5.2 Interpretation in relation to prior work

Explain where the findings agree with, extend or challenge the literature reviewed in Chapter 2. Consider credible alternative explanations. A strong discussion shows how the evidence supports the interpretation and where uncertainty remains.

5.3 Theoretical and practical implications

Separate implications that follow directly from the evidence from recommendations that depend on additional assumptions. Identify the intended audience and the conditions under which each implication is likely to apply.

5.4 Strengths and limitations

Discuss limitations precisely rather than adding a generic disclaimer. Explain their likely direction or consequence, the mitigation used, and what remains unresolved. Strengths should also be linked to specific design or evidence features.

Chapter 6

Conclusion

6.1 Research questions revisited

Answer each research question directly using findings already established in the thesis. The conclusion should not introduce a new dataset, method or major argument.

6.2 Contributions

Return to the claims in Table 1.1 and state the demonstrated contribution, its evidence and its boundary. Remove any contribution that the completed study does not actually substantiate.

6.3 Recommendations and future research

Prioritise a small number of feasible recommendations. Future research should arise from the study's limitations or new questions exposed by the findings, not from a generic statement that more research is needed.

6.4 Closing statement

End with a concise statement of what the reader should remember. This final paragraph is an opportunity to reconnect the technical contribution to the larger problem introduced in Chapter 1.

APPENDIX A

Example Research Instrument

[Replace this example: Replace this appendix with material required to understand or audit your study.]

Appendices may contain instruments, supplementary derivations, extended tables, technical specifications or other supporting evidence. Each appendix should be cited from the main thesis and should not be used to hide material essential to the argument.

A.1 Illustrative prompts

1. Describe *[Replace this example: the experience, process or observation]*.
2. What factors influenced *[Replace this example: the outcome of interest]*?
3. Is there anything else relevant to the research question?

Bibliography

Donald E. Knuth. *The TeXbook*. Addison-Wesley, 1984.

Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2 edition, 1994.